

Course: B. Tech

Branch: Common to all branches

Semester: III

Subject Code & Name: (BTBSC301) Engineering Mathematics - III

Max Marks: 60

Date: 05/02/2025

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The Laplace transform of $F(t) = \operatorname{erf}(\sqrt{t})$ is equal to	Understand CO1	1
	a. $\frac{1}{s\sqrt{s^2+1}}$ b. $\frac{1}{s\sqrt{s-1}}$ c. $\frac{1}{s\sqrt{s+1}}$ d. None		
2	The Laplace transform of $F(t) = e^{2t} t^3$ is equal to	Understand CO1	1
	a. $\frac{12}{(s-4)^2}$ b. $\frac{12}{(s+4)^2}$ c. $\frac{12}{(s+2)^4}$ d. None		
3	The Laplace transform of $f(t) = t^{-\frac{1}{2}}$ is equal to	Understand CO1	1
	a. $\frac{2as}{(s^2-a^2)^2}$ b. $\frac{2s}{(s^2-a^2)^2}$ c. $\frac{2as}{s^2-a^2}$ d. None		
4	The inverse Laplace transform of $\bar{f}(s) = \frac{1}{(s+3)^5}$ is equal to	Understand CO2	1
	a. $\frac{e^{-3t} t^4}{24}$ b. $\frac{e^{3t} t^4}{24}$ c. $e^{-3t} t^4$ d. None		
5	The inverse Laplace transform of $\bar{f}(s) = \frac{s^2-3s+4}{s^3}$ is equal to	Understand CO2	1
	a. $1-3t-2t^2$ b. $1+3t+2t^2$ c. $1-3t+2t^2$ d. None		
6	The inverse Laplace transform of the function $f(t) = \frac{1}{s^2+9}$ is	Understand CO2	1
	a. $\frac{1}{9}\sin 3t$ b. $\frac{1}{3}\sin 3t$ c. $\sin 3t$ d. None		
7	The Fourier cosine transform of the function $f(t)$ is	Understand CO3	1
	a. $F_c(s) = \int_0^\infty f(t) \cos st$ b. $F_c(s) = \int_0^\infty f(t) \cos t dt$ c. $F_c(s) = \int_0^\infty f(st) \cos t dt$ d. None		
8	The Fourier sine transform of e^{-ax} ($a > 0$) is	Understand CO3	1
	a. $\frac{s}{s^2+a^2}$ b. $\frac{a}{s^2+a^2}$ c. $\frac{s}{s^2-a^2}$ d. None		
9	The general solution of $zp = -x$ is given by	Understand CO4	1
	a. $\phi(x^2+z^2, y) = 0$ b. $\phi(x^2-z^2, y) = 0$ c. $\phi(x^2+z^2, 2y) = 0$ d. None		
10	The Lagrange's linear partial differential equation is of the form	Understand C04	1
	a. $Pp - Qq = R$ b. $Pp + Qq = 0$ c. $Pp + Qq = R$ d. None		

11	If $f(Z) = u + iv$ in Polar form is analytic then $\frac{\partial u}{\partial r}$ is equal to				Understand	1
	a. $\frac{\partial v}{\partial \theta}$	b. $r \frac{\partial v}{\partial \theta}$	c. $\frac{1}{r} \frac{\partial v}{\partial \theta}$	d. None	CO5	
12	A Bilinear transformation map circles into				Understand	1
	a. circles	b. hyperbola	c. parabola	d. None	CO5	
Q. 2	Solve the following.					12
A)	Find Laplace transform of $\frac{\cos 3t - \cos 2t}{t}$				Understand/ CO1	6
B)	Apply Laplace transform property to find Laplace transform of $\int_0^t te^{-4t} \sin 3t dt$				Apply/CO1	6
Q.3	Solve the following.					12
A)	Find inverse Laplace transform of $\log \left(\frac{s^2 + b^2}{s^2 + a^2} \right)$				Apply/CO2	6
B)	Solve $\frac{dy}{dt} + 2y = e^{-3t}$, $y(0) = 1$				Apply/CO2	6
Q. 4	Solve any TWO of the following.					12
A)	Find Fourier integral representation of the function $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$ and hence evaluate i) $\int_0^\infty \frac{\sin \omega \cos \omega x}{\omega} d\omega$ ii) $\int_0^\infty \frac{\sin \omega}{\omega} d\omega$				Apply/CO3	6
B)	Find the Fourier sine transform of $e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$.				Apply/CO3	6
C)	Using Parseval's identity, show that $\int_0^\infty \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$.				Apply/CO3	6
Q.5	Solve any TWO of the following.					12
A)	Obtain partial differential equation by eliminating arbitrary function f and g from $y = f(x - at) + g(x + at)$				Understand CO4	6
B)	Solve $yzp + zxq = xy$				Apply/CO4	6
C)	Use the method of separation of variables to solve the equation $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.				Apply /CO4	6
Q. 6	Solve any TWO of the following.					12
A)	Show that function $v = \sin x \cos y$ is harmonic function. Also find its harmonic conjugate function.				Remember /CO5	6
B)	Apply Cauchy's integral Formula to evaluate $\oint_C \frac{\sin^2 z}{(z - \frac{\pi}{2})^2} dz$ where $C: z = 1$				Apply/CO5	6
C)	State Cauchy's residue theorem and evaluate $\oint_C \frac{e^{-2z}}{(z-1)^2} dz$ where $C: z = 1.5$				Remember /CO5	6
	*****END*****					